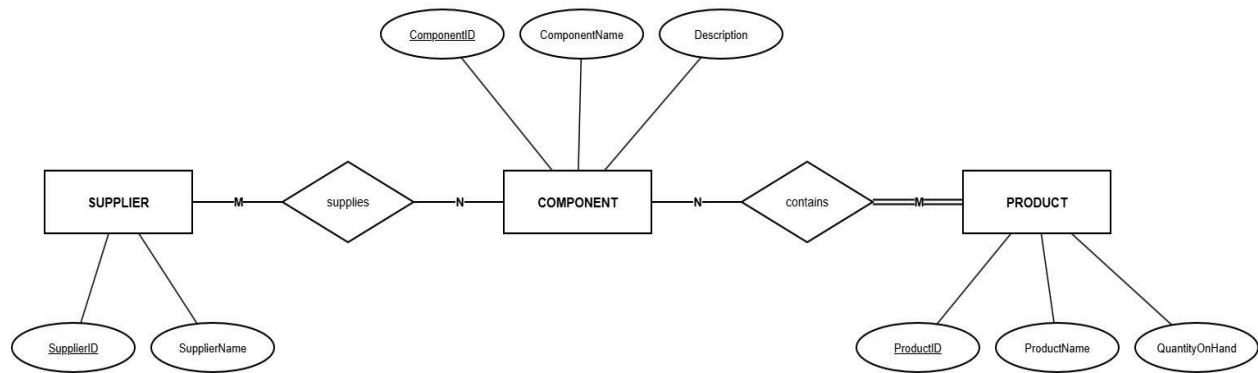


1.

Manufacturer ER Diagram (Chen notation)



Relational Schema:

SUPPLIER(SupplierID, SupplierName)

COMPONENT(ComponentID, ComponentName, Description)

PRODUCT(ProductID, ProductName, QuantityOnHand)

Since, "Supplies" & "contains" are many to many relationships separate tables are created for them:

SUPPLIES(SupplierID, ComponentID)

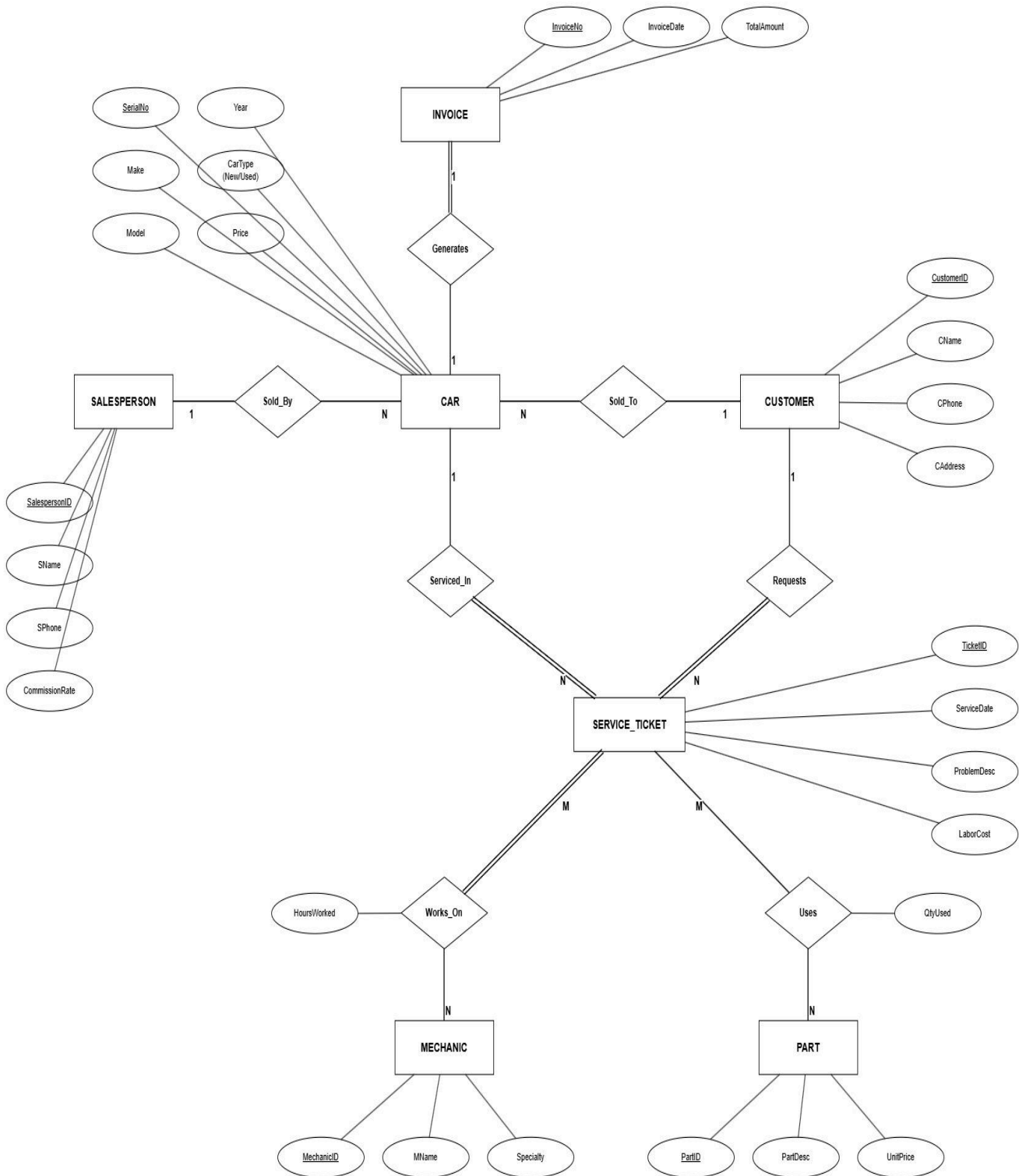
FOREIGN KEY (SupplierID) REFERENCES SUPPLIER(SupplierID),
FOREIGN KEY (ComponentID) REFERENCES COMPONENT(ComponentID)

CONTAINS (ComponentID, ProductID)

FOREIGN KEY (ComponentID) REFERENCES COMPONENT(ComponentID),
FOREIGN KEY (ProductID) REFERENCES PRODUCT(ProductID)

2.

Car Dealership - ER Diagram (Chen Notation)



SALESPERSON(SalespersonID, SName, SPhone, CommissionRate)

CUSTOMER(CustomerID, CName, CPhone, CAddress)

MECHANIC(MechanicID, MName, Specialty)

PART(PartID, PartDesc, UnitPrice, StockQty)

CAR(SerialNo, Make, Model, Year, CarType, Price, SaleDate, *SalespersonID*, *CustomerID*)

FOREIGN KEY (SalespersonID) REFERENCES SALESPERSON(SalespersonID),

FOREIGN KEY (CustomerID) REFERENCES CUSTOMER(CustomerID)

(Since "Sold_By" and "Sold_To" are 1:N relationships)

INVOICE(InvoiceNo, InvoiceDate, TotalAmount, *SerialNo*)

FOREIGN KEY (SerialNo) REFERENCES CAR(SerialNo),

SerialNo is NOT NULL and UNIQUE - exactly one invoice per car sale (1:1)

SERVICE_TICKET(TicketID, ServiceDate, ProblemDesc, LaborCost, *SerialNo*, *CustomerID*)

FOREIGN KEY (SerialNo) REFERENCES CAR(SerialNo),

FOREIGN KEY (CustomerID) REFERENCES CUSTOMER(CustomerID)

Since "Serviced_In" and "Requests" are 1:N relationships, SerialNo and CustomerID are posted into SERVICE_TICKET

Since "Works_On" and "Uses" are many-to-many relationships, separate tables are created for them:

WORKS_ON(TicketID, MechanicID, HoursWorked)

FOREIGN KEY (TicketID) REFERENCES SERVICE_TICKET(TicketID),
FOREIGN KEY (MechanicID) REFERENCES MECHANIC(MechanicID)

USES_PART(TicketID, PartID, QtyUsed, LineCost)

FOREIGN KEY (TicketID) REFERENCES SERVICE_TICKET(TicketID),
FOREIGN KEY (PartID) REFERENCES PART(PartID)